

Linear Algebra & Calculus.

Part A

1. Determine the rank of the matrix $A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 3 & -1 \\ 5 & 3 & -2 \end{bmatrix}$

Soln: $A = \begin{bmatrix} \textcircled{1} & -1 & 0 \\ 1 & 3 & -1 \\ 5 & 3 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & \textcircled{4} & -1 \\ 0 & 8 & -2 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 5R_1 \end{array}$

$$\sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 4 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 - \frac{8}{4} R_2$$

$\therefore \text{Rank of } A = 2$

2. What kind of conic section is represented by the quadratic form $7x_1^2 + 6x_1x_2 + 7x_2^2 = 200$. Transform it into canonical form.

Soln: $A = \begin{bmatrix} 7 & 3 \\ 3 & 7 \end{bmatrix}$

Characteristic equation is $\lambda^2 - \text{trace } A \lambda + |A| = 0$

$$\lambda^2 - 14\lambda + 40 = 0$$

$$(\lambda - 10)(\lambda - 4) = 0 \Rightarrow \lambda = 10, 4$$

$\therefore$ Eigen values are $\lambda = 4, 10$

Canonical form is $Q = \lambda_1 y_1^2 + \lambda_2 y_2^2$

$$Q = 4y_1^2 + 10y_2^2 = 200$$

$$\frac{y_1^2}{\left(\frac{200}{4}\right)} + \frac{y_2^2}{\left(\frac{200}{10}\right)} = 1, \text{ Ellipse.}$$

Part B

11 a) Test for consistency and solve the system of equations

$$x+2y-z=3, \quad 8x-y+2z=1, \quad 2x-2y+3z=2, \quad x-y+z=-1$$

Soln:

$$\tilde{A} = \begin{bmatrix} \textcircled{1} & 2 & -1 & : & 3 \\ 3 & -1 & 2 & : & 1 \\ 2 & -2 & 3 & : & 2 \\ 1 & -1 & 1 & : & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$R_4 \rightarrow R_4 - R_1$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & : & 3 \\ 0 & \textcircled{-7} & 5 & : & -8 \\ 0 & -6 & 5 & : & -4 \\ 0 & -3 & 2 & : & -4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{6}{7} R_2$$

$$R_4 \rightarrow R_4 - \frac{3}{7} R_2$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & : & 3 \\ 0 & -7 & 5 & : & -8 \\ 0 & 0 & \textcircled{5/7} & : & 20/7 \\ 0 & 0 & -1/7 & : & -4/7 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - \frac{-1}{5} R_3$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & : & 3 \\ 0 & -7 & 5 & : & -8 \\ 0 & 0 & 5/7 & : & 20/7 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

Rank $A = \text{Rank } \tilde{A} = \text{no. of unknowns} = 3$

$\therefore$ System of equations have unique solution.

By Back substitution,

$$\frac{5}{7} z = \frac{20}{7} \Rightarrow z = 4$$

$$-7y + 5z = -8 \Rightarrow 7y - 8 = 5z = 20 \Rightarrow y = 4$$

$$x + 2y - z = 3 \Rightarrow x + 2 \times 4 - 4 = 3$$

$$\Rightarrow x + 4 = 3 \Rightarrow x = -1$$

$\therefore$ solution is $x = -1, y = 4, z = 4$

11 b) Find the Eigen values and Eigen vectors of $\begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix}$

Characteristic equation is $\lambda^3 - \text{trace } A \lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$

$$\lambda^3 - 6\lambda^2 + (5+3+3)\lambda - 6 = 0$$

$$\lambda = 1, 2, 3$$

Eigen vector corresponding to $\lambda = 1$

$$\text{Consider } (A - \lambda I)x = 0$$

$$(A - I)x = 0$$

$$A - I = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\text{rank} = 2$$

$\therefore$ no. of eigen vectors corresponding to $\lambda = 1$

is no. of variables - rank = $3 - 2 = 1$

By Back substitution $x_2 + 2x_3 = 0$

Let $x_3 = 1$ $x_2 = -2$

$$-x_1 + x_2 + x_3 = 0$$

$$x_1 = x_2 + x_3 = -2 + 1 = -1$$

Eigen vector corresponding to $\lambda = 1$, $x_1 = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$

Eigen vector corresponding to $\lambda = 2$

Consider $(A - 2I)X = 0$

$$A - 2I = \begin{bmatrix} -1 & 1 & 2 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} -1 & 1 & 2 \\ 0 & -1 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$\sim \begin{bmatrix} -1 & 1 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$\text{Rank} = 2$$

$$\text{no. of Eigen vectors} = 3 - 2 = 1$$

By Back substitution,

$$-x_2 - x_3 = 0 \Rightarrow x_2 = -x_3$$

$$\text{Let } x_3 = 1 \therefore x_2 = -1$$

$$-x_1 + x_2 + 2x_3 = 0$$

$$x_1 = x_2 + 2x_3 = -1 + 2 = 1$$

Eigen vector corresponding to $\lambda = 2$

$$X_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Eigen vector corresponding to $\lambda = 3$

Consider $(A - 3I)X = 0$

$$A - 3I = \begin{bmatrix} -2 & 1 & 2 \\ -1 & -1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} -2 & 1 & 2 \\ 0 & -3/2 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad R_2 \rightarrow R_2 - \frac{1}{2} R_1$$

$$\sim \begin{bmatrix} -2 & 1 & 2 \\ 0 & -3/2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 + \frac{2}{3} R_2$$

$$\text{Rank} = 2$$

$$\text{No. of eigen vectors} = 3 - 2 = 1$$

By Back substitution,

$$-\frac{3}{2} x_2 = 0 \Rightarrow x_2 = 0$$

$$-2x_1 + x_2 + 2x_3 = 0$$

$$\text{Let } x_3 = 1$$

$$2x_1 = x_2 + 2x_3 = 0 + 2 = 2$$

$$x_1 = 1$$

Eigen vector corresponding to $\lambda = 3$

$$X_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

12 a) For what value of a and b the system of equations $x+y+z=6$, $x+2y+3z=10$, $x+2y+az=b$ have 1) no solution 2) unique solution 3) more than one solution.

Soln: $\tilde{A} = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 2 & 3 & : & 10 \\ 1 & 2 & a & : & b \end{bmatrix}$

$$\sim \begin{bmatrix} \textcircled{1} & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 1 & a-1 & : & b-6 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & \textcircled{1} & 2 & : & 4 \\ 0 & 0 & a-3 & : & b-10 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

No solution when $\text{Rank } A \neq \text{Rank } \tilde{A}$

when $a=3$ and $b \neq 10$, $\text{Rank } A = 2$, $\text{Rank } \tilde{A} = 3$

$\therefore$ No solution when $a=3$, $b \neq 10$

Unique solution when $\text{Rank } A = \text{Rank } \tilde{A} = 3$

i.e. when $a \neq 3$ and b may have any value

Infinite no. of solutions when $\text{Rank } A = \text{Rank } \tilde{A} \neq 3$

i.e. when $a=3$ and $b=10$, $\text{Rank } A = \text{Rank } \tilde{A} = 2$.

12 b) Find the matrix of transformation that diagonalize the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$. Also write the diagonal matrix.

Soln:

Characteristic equation is $\lambda^3 - \text{trace } A \lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$

$$\lambda^3 - 18\lambda^2 + [5 + 20 + 20]\lambda - 0 = 0$$

Eigen values of are $\lambda = 0, 3, 15$

Eigen vector corresponding to $\lambda = 0$

Consider $(A - \lambda I)x = 0$

$$A - 0I = A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 8 & -6 & 2 \\ 0 & \frac{20}{8} & -\frac{10}{4} \\ 0 & -\frac{10}{4} & \frac{5}{2} \end{bmatrix}$$

$$R_2 \rightarrow R_2 \rightarrow R_2 + \frac{6}{8}R_1$$

$$R_3 \rightarrow R_3 - \frac{2}{8}R_1$$

$$\sim \begin{bmatrix} 8 & -6 & 2 \\ 0 & \frac{20}{8} & -\frac{10}{4} \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

By Back substitution $\frac{20}{8}x_2 - \frac{10}{4}x_3 = 0$

$$x_2 = x_3$$

$$\text{Let } x_2 = 2 \Rightarrow x_3 = 2$$

$$8x_1 - 6x_2 + 2x_3 = 0$$

$$8x_1 = 6x_2 - 2x_3 = 6(2) - 2(2) \Rightarrow x_1 = \frac{4}{2} = 2$$

Eigen vector corresponding to $\lambda = 0$

$$X_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Eigen vector corresponding to $\lambda = 3$

Consider $(A - \lambda I)X = 0$

$$A - \lambda I = \begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 5 & -6 & 2 \\ 0 & -16/5 & -8/5 \\ 0 & -8/5 & -4/5 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - \frac{-6}{5} R_1$$

$$R_3 \rightarrow R_3 - \frac{2}{5} R_1$$

$$\sim \begin{bmatrix} 5 & -6 & 2 \\ 0 & -16/5 & -8/5 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{1}{2} R_2$$

By Back substitution $-\frac{16}{5}x_2 - \frac{8}{5}x_3 = 0$

$$-2x_2 = x_3$$

$$\text{Let } x_2 = 1$$

$$x_3 = -2$$

$$5x_1 - 6x_2 + 2x_3 = 0$$

$$5x_1 = 6x_2 - 2x_3 = 6 \times 1 - 2 \times (-2) = 10$$

$$x_1 = 2$$

Eigen vector corresponding to $\lambda = 3$

$$X_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

Eigen vector corresponding to $\lambda = 15$

Consider $(A - 15I)x = 0$

$$A - 15I = \begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix}$$

$$\sim \begin{bmatrix} -7 & -6 & 2 \\ 0 & -\frac{20}{7} & -\frac{40}{7} \\ 0 & -\frac{40}{7} & -\frac{80}{7} \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - \frac{6}{7}R_1 \\ R_3 \rightarrow R_3 + \frac{2}{7}R_1 \end{array}$$

$$\sim \begin{bmatrix} -7 & -6 & 2 \\ 0 & -\frac{20}{7} & -\frac{40}{7} \\ 0 & 0 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 - 2R_2$$

By Back substitution,

$$-\frac{20}{7}x_2 - \frac{40}{7}x_3 = 0$$

$$-x_2 - 2x_3 = 0 \Rightarrow x_2 = -2x_3$$

$$\text{Let } x_3 = 1$$

$$x_2 = -2$$

$$-7x_1 - 6x_2 + 2x_3 = 0$$

$$7x_1 = -6x_2 + 2x_3 = -6(-2) + 2(1) = 14$$

$$x_1 = 2$$

Eigen vector corresponding to $\lambda = 15$

$$X_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

$\therefore$ Matrix of transformation that diagonalize the matrix

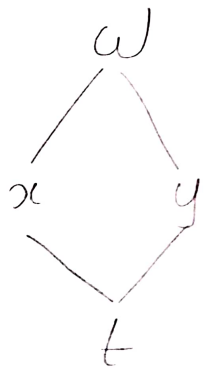
$$P = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

Diagonal matrix

$$D = P^{-1}AP = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix}$$

83. Find the derivative of $w = x^2 + y^2$ w.r to t along the path $x = at^2$, $y = 2at$.

$$\begin{aligned}\frac{dw}{dt} &= \frac{\partial w}{\partial x} \times \frac{dx}{dt} + \frac{\partial w}{\partial y} \times \frac{dy}{dt} \\&= 2x \times 2at + 2y \times 2a \\&= 2at^2 \times 2at + 2 \times 2at + 2a \\&= \underline{\underline{4a^2t^3 + 8a^2t}}\end{aligned}$$



Module 2

4. Let $f(x, y) = \sqrt{3x + 2y}$, find the slope of the surface $z = f(x, y)$ in the y -direction at the point $(2, 5)$.

Ans:
$$\frac{\partial z}{\partial y} = \frac{1}{2\sqrt{3x+2y}} \times 2 = \frac{1}{\sqrt{3x+2y}}$$

$$\begin{aligned} \left(\frac{\partial z}{\partial y} \right)_{(2,5)} &= \frac{1}{\sqrt{3(2)+2(5)}} = \frac{1}{\sqrt{6+10}} \\ &= \frac{1}{\sqrt{16}} = \frac{1}{4} // \end{aligned}$$

Slope of the surface in y -direction @ $(2, 5)$.

¹³
Q. a) If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ find the value of x

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$$

a) Let $a = \frac{x}{y}$, $b = \frac{y}{z}$, $c = \frac{z}{x}$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial a} \times \frac{\partial a}{\partial x} + \frac{\partial u}{\partial c} \times \frac{\partial c}{\partial x}$$

$$= \frac{\partial u}{\partial a} \times \frac{1}{y} + \frac{\partial u}{\partial c} \times \frac{-z}{x^2}$$

$$= \frac{\partial u}{\partial a} \times \frac{1}{y} - \frac{\partial u}{\partial c} \times \frac{z}{x^2} \longrightarrow \textcircled{1}$$



$$\begin{aligned}
 \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial a} \times \frac{\partial a}{\partial y} + \frac{\partial u}{\partial b} \times \frac{\partial b}{\partial y} \\
 &= \frac{\partial u}{\partial a} \times \frac{-x}{y^2} + \frac{\partial u}{\partial b} \times \frac{1}{z} \\
 &= -\frac{\partial u}{\partial a} \frac{x}{y^2} + \frac{\partial u}{\partial b} \times \frac{1}{z} \longrightarrow (2)
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial u}{\partial z} &= \frac{\partial u}{\partial b} \times \frac{\partial b}{\partial z} + \frac{\partial u}{\partial c} \times \frac{\partial c}{\partial z} \\
 &= -\frac{y}{z^2} \frac{\partial u}{\partial b} + \frac{\partial u}{\partial c} \times \frac{1}{x} \longrightarrow (3)
 \end{aligned}$$

$$\begin{aligned}
 x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} + z \cdot \frac{\partial u}{\partial z} \\
 = 0
 \end{aligned}$$

$$\begin{aligned}
 \text{ie, } &\left[x \frac{\partial u}{\partial a} \times \frac{1}{y} - \frac{\partial u}{\partial c} \times \frac{z}{x} \right] + \left[-\frac{\partial u}{\partial a} \frac{x}{y} + \frac{\partial u}{\partial b} \frac{y}{z} \right] \\
 &+ \left[-\frac{y}{z} \frac{\partial u}{\partial b} + \frac{\partial u}{\partial c} \frac{z}{x} \right] \\
 &= \underline{\underline{0}}
 \end{aligned}$$

13 b) If the local linear approximation of a function

$$f(x, y, z) = xy + z^2 \text{ at a point } P \text{ is } L(x, y, z) = y + 2z - x,$$

find the point P .

Ans:

$$L(x, y, z) = f(x_0, y_0, z_0) + f_x(x_0, y_0, z_0)(x - x_0) \\ + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0)$$

$$f(x_0, y_0, z_0) = x_0 y_0 + z_0^2$$

$$f_x = y \quad ; \quad f_x(x_0, y_0, z_0) = y_0$$

$$f_y = x \quad ; \quad f_y(x_0, y_0, z_0) = x_0$$

$$f_z = 2z \quad ; \quad f_z(x_0, y_0, z_0) = 2z_0$$

$$L(x, y, z) = x_0 y_0 + z_0^2 + y_0(x - x_0) + x_0(y - y_0) + 2z_0(z - z_0)$$

$$y + 2z - x = x_0 y_0 + z_0^2 + y_0 x - y_0 x_0 + x_0 y - x_0 y_0 + 2z_0 z - 2z_0^2$$

$$y + 2z - x = \cancel{y_0 x} \quad x_0 y + 2z_0 z + y_0 x + x_0 y_0 + z_0^2 - 2z_0^2$$

Comparing coefficients of x, y, z .

$$y = x_0 y \quad \Rightarrow \quad x_0 = 1$$

$$2z = 2z_0 z \quad \Rightarrow \quad z_0 = 1$$

$$-x = y_0 x \quad \Rightarrow \quad y_0 = -1$$

$$\therefore P(x_0, y_0, z_0) = (1, -1, 1)$$

14. a) If $z = e^{xy}$, $x = 2u + v$, $y = \frac{u}{v}$ find $\frac{\partial z}{\partial u}$

b) Locate all relative extrema of $f(x, y) = 3x^2 - 2xy + y^2 - 8y$.

$$\begin{aligned}
 a) \quad \frac{\partial z}{\partial u} &= \frac{\partial z}{\partial x} \times \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \times \frac{\partial y}{\partial u} \\
 &= \left[e^{xy} \times y \right] \times 2 + \left[e^{xy} \times x \right] \times \frac{1}{v} \\
 &= 2ye^{xy} + \frac{x}{v} e^{xy} \\
 &= 2x \frac{u}{v} e^{(2u+v) \times \frac{u}{v}} + \frac{2u+v}{v} \times e^{((2u+v) \times \frac{u}{v})} \\
 &= e^{(2u+v) \times \frac{u}{v}} \left[\frac{2u}{v} + \frac{2u+v}{v} \right]
 \end{aligned}$$

b) Critical points

$$f_x = 6x - 2y$$

$$f_y = -2x + 2y - 8$$

$$f_x = 0$$

$$\Rightarrow 6x - 2y = 0 \longrightarrow (1)$$

$$f_y = 0$$

$$\Rightarrow -2x + 2y - 8 = 0 \longrightarrow (2)$$

$$(1) + (2)$$

$$= 4x - 8 = 0$$

$$\therefore x = 2$$

$$y = 6$$

The critical point is
 $= (2, 6)$

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

$$f_{xx} = 6$$

$$f_{yy} = 2$$

$$f_{xy} = -2$$

$$\therefore D = (6 \times 2) - 4$$
$$= 8$$

$$D > 0$$

$$f_{xx} = 6 > 0$$

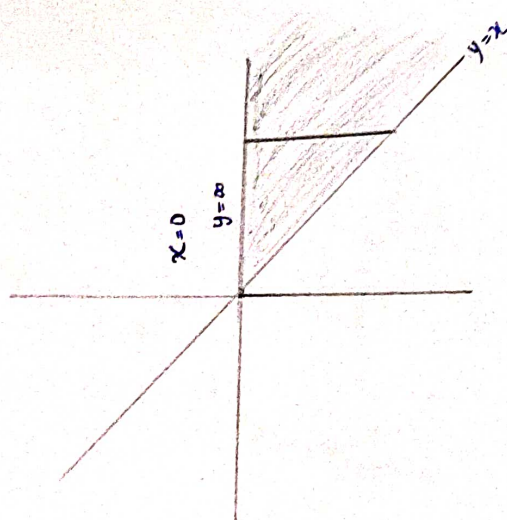
$\therefore F$ has relative minimum at $(2, 6)$

Module - 3

15. @ Evaluate $\int_0^\infty \int_x^\infty e^{-y/y} dy dx$ by reversing order of integration

$$A) \int_0^\infty \int_0^\infty \frac{e^{-y}}{y} dy dx$$

Given $y = x$; $y = \infty$
 $x \rightarrow 0$ to y
 $y \rightarrow 0$ to ∞



∴ Order of Integration now is

$$\int_0^\infty \int_0^y \frac{e^{-y}}{y} dx dy$$

$$\left[\int_b^a e^{-y} = [e^{-y}]_b^a \right]$$

$$= \int_0^\infty \frac{e^{-y}}{y} [x]_0^y dy \Rightarrow \int_0^\infty \frac{e^{-y}}{y} [y] dy$$

$$\Rightarrow \int_0^\infty e^{-y} dy \Rightarrow [e^{-y}]_0^\infty \Rightarrow [e^{-\infty} - e^{-0}]$$

$$\Rightarrow [0 + 1] = \underline{\underline{1}}$$

15. b) Find the volume of the solid in the first octant bounded by the coordinate planes and the plane $x+2y+z=6$

ans.)

$$\text{Volume} = \int \int \int dV$$

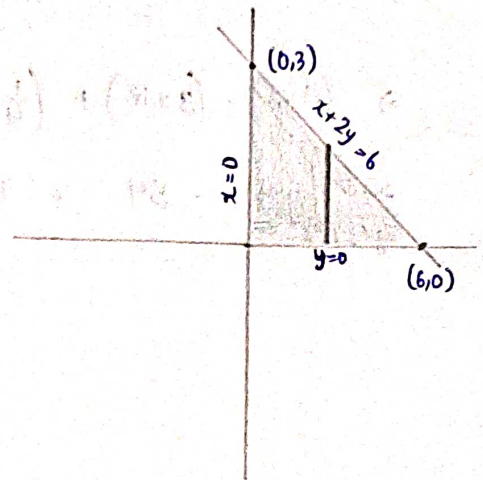
$$z \rightarrow 0 \text{ to } 6-x-2y$$

$$x+2y=6$$

$$\rightarrow \begin{array}{|c|c|c|} \hline x & 0 & 6 \\ \hline y & 3 & 0 \\ \hline \end{array}$$

The points are (0,3) (6,0)

$$y \rightarrow 0 \text{ to } 3-x/2 ; \quad x \rightarrow 0 \text{ to } 6$$



$$\int_0^6 \int_0^{3-\frac{x}{2}} \int_0^{6-2y-x} dz dy dx$$

$$\Rightarrow \int_0^6 \int_0^{3-\frac{x}{2}} [6-2y-x] dy dx$$

$$\Rightarrow \int_0^6 \left[6 \left[y \right]_0^{3-\frac{x}{2}} - 2 \left[\frac{y^2}{2} \right]_0^{3-\frac{x}{2}} - x \left[y \right]_0^{3-\frac{x}{2}} \right] dx$$

$$\Rightarrow \int_0^6 \left[6 \left(3-\frac{x}{2} \right) - 2 \left(3-\frac{x}{2} \right) - \frac{2}{2} \left[9 - 3\frac{x}{2} + \frac{x^2}{4} \right] \right] dx$$

$$\Rightarrow \int_0^6 \left[18 - \frac{6x}{2} - 3x + \frac{x^2}{2} - 9 + 3x - \frac{x^2}{4} \right] dx$$

$$\Rightarrow \int_0^6 9 - 3x + \frac{x^2}{4} dx$$

$$\Rightarrow \left[9[x]_0^6 - 3\left[\frac{x^2}{2}\right]_0^6 + \frac{1}{4}\left[\frac{x^3}{3}\right]_0^6 \right]$$

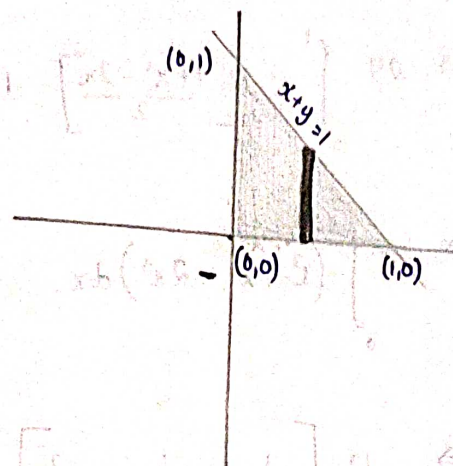
$$\Rightarrow (9 \times 6) - (3 \times 18) + \left(\frac{1}{4} \times 72\right)$$

$$\Rightarrow 54 - 54 + 18$$

$$\Rightarrow \underline{18}.$$

16. (a) Find the mass and centre of gravity of the triangular lamina with vertices $(0,0)$, $(0,1)$, $(1,0)$ with density function $\delta(x,y) = xy$.

$$A) M = \iint_R \delta(x,y) dA$$



We have $\delta(x,y) = xy$
 $y \rightarrow 0 \text{ to } 1-x$
 $x \rightarrow 0 \text{ to } 1$

$$M = \int_0^1 \int_0^{1-x} xy \, dy \, dx$$

$$= \int_0^1 x \left[\frac{y^2}{2} \right]_0^{1-x} dx \Rightarrow \int_0^1 x \left[\frac{1+x^2-2x}{2} \right] dx$$

$$\Rightarrow \int_0^1 \frac{1}{2} x [1+x^2-2x] dx \Rightarrow \frac{1}{2} \int_0^1 x + x^3 - 2x^2 dx$$

$$\Rightarrow \frac{1}{2} \left[\left[\frac{x^2}{2} \right]_0^1 + \left[\frac{x^4}{4} \right]_0^1 - 2 \left[\frac{x^3}{3} \right]_0^1 \right] \Rightarrow \frac{1}{2} \left[\frac{1}{2} + \frac{1}{4} - \frac{2}{3} \right]$$

$$\Rightarrow \frac{1}{2} \left[\frac{1}{12} \right] \Rightarrow \frac{1}{24}$$

Centre of gravity $= \bar{x} = \frac{1}{M} \cdot \iint_R x \delta(x,y) dA$

We have $M = \frac{1}{24}$,

$$\Rightarrow \frac{1}{24} \iint_R x(xy) dA$$

$$\Rightarrow 24 \int_0^1 \int_0^{1-x} x^2 y dy dx \Rightarrow 24 \int_0^1 x^2 \left[\frac{y^2}{2} \right]_0^{1-x} dx$$

$$\Rightarrow 24 \int_0^1 x^2 \left[\frac{1+x^2-2x}{2} \right] dx \Rightarrow \frac{24}{2} \int_0^1 x^2 (1+x^2-2x) dx$$

$$\Rightarrow 12 \int_0^1 (x^2 + x^4 - 2x^3) dx \Rightarrow 12 \left[\left[\frac{x^3}{3} \right]_0^1 + \left[\frac{x^5}{5} \right]_0^1 - 2 \left[\frac{x^4}{4} \right]_0^1 \right]$$

$$\Rightarrow 12 \left[\frac{1}{3} + \frac{1}{5} - \frac{2}{4} \right] \Rightarrow 12 \times \left[-\frac{1}{30} \right]$$

$$\Rightarrow \frac{12}{30} = \frac{2}{5}$$

Centre of gravity $\bar{y} = \frac{1}{M} \iint_R y \delta(x,y) dA$

$$= 24 \int_0^1 \int_0^{1-x} xy^2 dy dx \Rightarrow 24 \int_0^1 x \left[\frac{y^3}{3} \right]_0^{1-x} dx$$

$$\Rightarrow 24 \int_0^1 x \left[\frac{1-3x+3x^2-x^3}{3} \right] dx$$

$$\Rightarrow \frac{24}{3} \int_0^1 x (1-3x+3x^2-x^3) dx \Rightarrow 8 \int_0^1 (x-3x^2+3x^3-x^4) dx$$

$$\Rightarrow 8 \left[\left[\frac{x^2}{2} \right]_0^1 - 3 \left[\frac{x^3}{3} \right]_0^1 + 3 \left[\frac{x^4}{4} \right]_0^1 - \left[\frac{x^5}{5} \right]_0^1 \right] \Rightarrow 8 \left[\frac{1}{2} - \frac{3}{3} + \frac{3}{4} - \frac{1}{5} \right]$$

$$\Rightarrow 8 \times \frac{1}{20} = \frac{2}{5}$$

16. b) Evaluate $\iint_R x^2 dy dx$ where R is the region between $y=x$ and $y=x^2$.

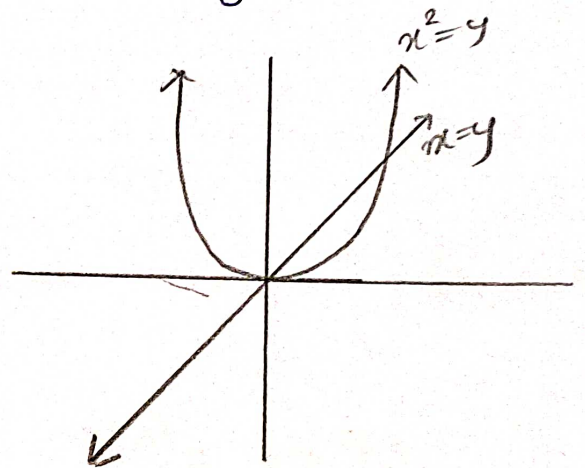
ans:- $\int_0^1 \int_{x^2}^x x^2 dy dx$

$$= \int_0^1 x^2 [y]_{x^2}^x dx$$

$$= \int_0^1 x^2 (x - x^2) dx$$

$$= \left[\frac{x^4}{4} \right]_0^1 - \left[\frac{x^5}{5} \right]_0^1$$

$$= \frac{1}{4} - \frac{1}{5} = \underline{\underline{\frac{1}{20}}}$$



5. Evaluate $\int_0^a \int_0^a \int_0^a (yz + xz + xy) dx dy dz$

ans:- $\int_0^a \int_0^a \int_0^a (yz + xz + xy) dx dy dz = \int_0^a \int_0^a \left[yz \left[x \right]_0^a + \left[\frac{x^2}{2} \right]_0^a z + \left[\frac{xy^2}{2} \right]_0^a \right] dy dz$

$$= \int_0^a \left[a \cdot z \left[\frac{y^2}{2} \right]_0^a + \frac{a^2}{2} \cdot z \cdot [y]_0^a + \frac{a^2}{2} \cdot \left[\frac{y^2}{2} \right]_0^a \right] dz$$

$$= \int_0^a \left[a \cdot \frac{a^2}{2} \cdot \left[\frac{z^2}{2} \right]_0^a + \left[a \cdot \frac{a^2}{2} \cdot \left[\frac{z^2}{2} \right]_0^a \right] + \left[\frac{a^2}{2} \cdot \frac{a^2}{2} \cdot [z]_0^a \right] \right] dz$$

$$= a \cdot \frac{a^2}{2} + \frac{a^2}{2} + a \cdot \frac{a^2}{2} \cdot \frac{a^2}{2} + a \cdot \frac{a^2}{2} \cdot \frac{a^2}{2} = \underline{\underline{\frac{3a^5}{4}}}$$

6. Use Polar co-ordinates to evaluate $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{3/2} dy dx$

ans:- $x = -1, x = 1, y = 0, y = \sqrt{1-x^2} \Rightarrow x^2 + y^2 = 1$

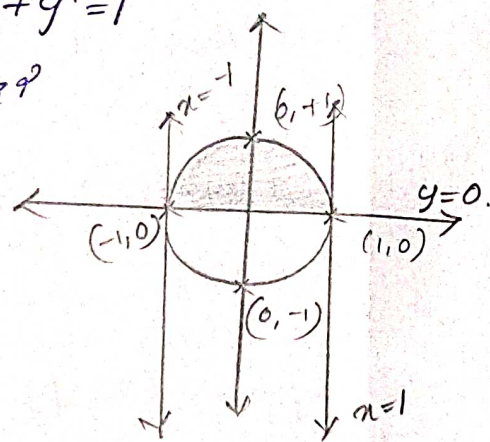
Substitute $x = r \cos \theta, y = r \sin \theta, x^2 + y^2 = r^2$

$dy dx = r dr d\theta$

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{3/2} dy dx = \int_0^1 \int_0^\pi (r^2)^{3/2} \cdot r d\theta dr$$

$$= \int_0^1 \int_0^\pi r^4 d\theta dr = \int_0^1 r^4 [0]_0^\pi dr$$

$$= \pi \cdot \left[\frac{r^5}{5} \right]_0^1 = \underline{\underline{\frac{\pi}{5}}}$$



$r: 0 \rightarrow 1$

$\theta: 0 \rightarrow \pi$

PART-A

7. Test the convergence of $\sum_{k=1}^{\infty} \frac{1}{(2k+3)^{17}}$

Ans: Consider the series $\sum b_k = \sum \frac{1}{k^{17}}$ which is a convergent p-series,

$$p = \lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \lim_{k \rightarrow \infty} \frac{1}{(2k+3)^{17}} \cdot k^{17}$$

$$= \lim_{k \rightarrow \infty} \frac{k^{17}}{k^{17} \left(2 + \frac{3}{k}\right)^{17}}$$

$$= \frac{1}{2} > 0 \neq \text{finite.}$$

by limit comparison test, since $\sum_{k=1}^{\infty} \frac{1}{k^{17}}$ converges, $\sum_{k=1}^{\infty} \frac{1}{(2k+3)^{17}}$ also converges.

8. Examine whether the series converges or not.

$$\sum_{k=1}^{\infty} \frac{1}{(\ln(k+1))^k}$$

Ans: By root test, $p = \lim_{k \rightarrow \infty} (a_k)^{1/k}$

$$= \lim_{k \rightarrow \infty} \left(\frac{1}{(\ln(k+1))^k} \right)^{1/k}$$

$$= \lim_{k \rightarrow \infty} \frac{1}{\ln(k+1)} = 0 < 1$$

$\therefore$ the series converges.

Module-IV

17a) Discuss the convergence of the series

i) $\sum_{k=1}^{\infty} \frac{k!}{k^k}$

ii) $\sum_{k=1}^{\infty} \left(\frac{k}{k+1}\right)^{k^2}$

Ans:

$$a_k = \frac{k!}{k^k}, \quad a_{k+1} = \frac{(k+1)!}{(k+1)^{k+1}}$$

$$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{(k+1)!}{(k+1)^{k+1}} \cdot \frac{k^k}{k!}$$

$$= \lim_{k \rightarrow \infty} \frac{(k+1) \cdot k^k}{(k+1)^k \cdot (k+1)} = \lim_{k \rightarrow \infty} \left(\frac{k}{k+1}\right)^k$$

$$= \lim_{k \rightarrow \infty} \left(\frac{k}{k(1+\frac{1}{k})}\right)^k = \lim_{k \rightarrow \infty} \frac{1}{\left(1+\frac{1}{k}\right)^k} = \frac{1}{e} < 1$$

$\therefore$ by ratio test $\sum a_k$ converges.

ii) Let $a_k = \left(\frac{k}{k+1}\right)^{k^2}$

$$\lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} \left[\left(\frac{k}{k+1}\right)^{k^2} \right]^{\frac{1}{k}} = \lim_{k \rightarrow \infty} \left(\frac{k}{k+1}\right)^k$$

$$= \lim_{k \rightarrow \infty} \left(\frac{1}{1+\frac{1}{k}}\right)^k = \frac{1}{e} < 1$$

$\therefore$ by root test the series converges.

17 b) Examine the convergence and divergence of the series $\frac{x}{1 \cdot 3} + \frac{x^2}{3 \cdot 5} + \frac{x^3}{5 \cdot 7} + \dots$

Ans: Let $a_k = \frac{x^k}{(2k-1)(2k+1)}$

$$a_{k+1} = \frac{x^{k+1}}{(2(k+1)-1)(2(k+1)+1)} = \frac{x^{k+1}}{(2k+1)(2k+3)}$$

$$\begin{aligned} \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} &= \lim_{k \rightarrow \infty} \frac{x^{k+1}}{(2k+1)(2k+3)} \cdot \frac{(2k-1)(2k+1)}{x^k} \\ &= \lim_{k \rightarrow \infty} \frac{x(2k-1)}{(2k+3)} = x \lim_{k \rightarrow \infty} \frac{2k(1 - \frac{1}{2k})}{2k(1 + \frac{3}{2k})} \\ &= x \end{aligned}$$

If $x < 1$, the series converges, and $x > 1$ the series diverges:

If $x = 1$, the series becomes, $\sum \frac{1}{(2k-1)(2k+1)}$

$$\text{Let } a_k = \frac{1}{(2k-1)(2k+1)} = \frac{1}{4k^2 - 1}$$

$$\text{Let } b_k = \frac{1}{k^2}$$

$$\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \lim_{k \rightarrow \infty} \frac{k^2}{4k^2 - 1} = \frac{1}{4} > 0 \text{ \& finite}$$

$\therefore$ by limit comparison test, since $\sum \frac{1}{k^2}$ is a convergent p-series $\sum a_k$ is also convergent.

18a) Test the convergence of $1 + \frac{1 \cdot 3}{3!} + \frac{1 \cdot 3 \cdot 5}{5!} + \frac{1 \cdot 3 \cdot 5 \cdot 7}{7!} + \dots$

Ans: Let $a_k = \frac{1 \cdot 3 \cdot 5 \cdot 7 \dots (2k-1)}{(2k-1)!}$

$$a_{k+1} = \frac{1 \cdot 3 \cdot 5 \cdot 7 \dots (2(k+1)-1)}{(2(k+1)-1)!}$$

$$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{1 \cdot 3 \cdot 5 \dots (2k+1)}{(2k+1)!} \cdot \frac{(2k-1)!}{1 \cdot 3 \cdot 5 \dots (2k-1)}$$

$$= \lim_{k \rightarrow \infty} \frac{(2k+1)}{2k(2k+1)} = 0 < 1$$

$\therefore$ the series converges by ratio test.

18b) Prove that the series $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{(k-2)}{k(k+1)}$ is conditionally convergent.

Ans: Let $a_k = \frac{k-2}{k(k+1)}$

$$a_{k+1} = \frac{(k+1)-2}{(k+1)(k+2)} = \frac{k-1}{(k+1)(k+2)}$$

$$\therefore a_k > a_{k+1}$$

$$\lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} \frac{k-2}{k^2(1+\frac{1}{k})} = 0$$

$\therefore$ by Leibnitz test the series converges.

Consider $|a_k| = \frac{k-2}{k(k+1)}$

Let $b_k = \frac{1}{k}$

$$\begin{aligned}\lim_{k \rightarrow \infty} \frac{a_k}{b_k} &= \lim_{k \rightarrow \infty} \frac{k-2}{k^2+k} \cdot k = \lim_{k \rightarrow \infty} \frac{k^2-2k}{k^2+k} \\ &= \lim_{k \rightarrow \infty} \frac{k^2 \left(1 - \frac{2}{k}\right)}{k^2 \left(1 + \frac{1}{k}\right)} \\ &= 1 > 0 \text{ \& finite.}\end{aligned}$$

Since $\sum b_k$ is a divergent p-series, by limit comparison test $\sum |a_k|$ is also divergent.

~~$\sum a_k$ converges & $\sum |a_k|$ diverges~~

$\sum |a_k|$ diverges $\Rightarrow \sum a_k$ diverges absolutely.

Since $\sum a_k$ converges & diverges absolutely

$\Rightarrow \sum a_k$ is conditionally convergent.

PART A

9. Find the Maclaurin series of $\frac{1}{x+1}$ upto third degree term.

Soln Maclaurin series of $f(x)$ is

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$\text{Given } f(x) = \frac{1}{x+1}$$

$$f(0) = \frac{1}{0+1} = 1$$

$$f'(x) = \frac{-1}{(1+x)^2}$$

$$f'(0) = \frac{-1}{1^2} = -1$$

$$f''(x) = \frac{+2}{(1+x)^3}$$

$$f''(0) = \frac{+2}{1^3} = +2$$

$$\begin{aligned} f'''(x) &= +2x - 3x(1+x)^{-4} \\ &= \frac{-6}{(1+x)^4} \end{aligned}$$

$$f'''(0) = -6$$

$$\begin{aligned} \therefore f(x) &= 1 + x(-1) + \frac{x^2}{2} \times 2 + \frac{x^3}{6} \times (-6) + \dots \\ &= 1 - x + x^2 - x^3 + \dots \end{aligned}$$

==

10. Find the half range sine series of $f(x) = x$ in $0 < x < \pi$

Soln: Half range sine series is,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \sin(n\pi x / \pi) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \sin(nx) dx$$

$$= \frac{2}{\pi} \left[x \times \frac{\cos(nx)}{n} - 1 \times \frac{-1}{n} \frac{\sin(nx)}{n} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\frac{-\pi}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) - 0 \right]$$

$$= \frac{2}{\pi} \left[\frac{-\pi}{n} \times (-1)^n \right]$$

$$= \frac{-2}{n} (-1)^n$$

$$f(x) = \sum_{n=1}^{\infty} \underline{\underline{\frac{-2}{n} (-1)^n \sin(nx)}}$$

19 a) Obtain the Fourier series for the function

$$f(x) = |\sin x| \text{ in } -\pi < x < \pi$$

Soln: Since $f(x)$ is even, $b_n = 0$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

$$a_0 = \frac{1}{2L} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{2\pi} \times 2 \int_0^{\pi} |\sin x| dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin x dx$$

$$= \frac{1}{\pi} [-\cos x]_0^{\pi}$$

$$= \frac{1}{\pi} [-\cos \pi - (-\cos 0)]$$

$$= \frac{1}{\pi} [-1 + 1] = \frac{2}{\pi}$$

$$a_n = \frac{1}{L} \int_{-\pi}^{\pi} f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} |\sin x| \cos(nx) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \sin x \cos(nx) dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin(1+n)x + \sin(1-n)x dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin(n+1)x - \sin(n-1)x dx$$

$$= \frac{1}{\pi} \left[\frac{-\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{-\cos(n+1)\pi}{n+1} + \frac{\cos(n-1)\pi}{n-1} - \left[\frac{-\cos 0}{n+1} - \frac{\cos 0}{n-1} \right] \right]$$

$$= \frac{1}{\pi} \times \frac{2}{n^2-1} [(-1)^{n-1} - 1] \quad \text{if } n \neq 1$$

$$\boxed{\cos(n-1)\pi = (-1)^{n-1}}$$

When $n=1$, $a_1 = \frac{2}{\pi} \int_0^{\pi} \sin x \cos x \, dx$

$$= \frac{2}{\pi} \int_0^{\pi} \frac{\sin 2x}{2} \, dx$$

$$= \frac{2}{\pi} \times \left[-\frac{\cos 2x}{4} \right]_0^{\pi}$$

$$= \left[-\frac{\cos 2x}{\pi 2} \right]_0^{\pi}$$

$$= \frac{-1}{2\pi} [\cos 2\pi - \cos 0]$$

$$= \frac{-1}{2\pi} [1 - 1]$$

$$= \underline{\underline{0}}$$

19 b) If $f(x) = \begin{cases} kx, & 0 < x < \pi/2 \\ k(\pi-x), & \pi/2 < x < \pi \end{cases}$ then show that

$$f(x) = \frac{4k}{\pi} \left[\frac{\sin x}{1^2} - \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} - \dots \right]$$

Soln: From the question we see that $f(x)$ has to be expanded as a half range Fourier sine series in $(0, \pi)$.

Fourier sine series of $f(x)$ is,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

$$= \frac{2}{\pi} \left[\int_0^{\pi/2} kx \sin nx \, dx + \int_{\pi/2}^{\pi} k(\pi-x) \sin nx \, dx \right]$$

$$= \frac{2}{\pi} \left\{ \left[kx \left[-\frac{\cos nx}{n} \right] - k \left[-\frac{\sin nx}{n^2} \right] \right]_0^{\pi/2} + \left[k(\pi-x) \left[-\frac{\cos nx}{n} \right] - (-k) \left[-\frac{\sin nx}{n^2} \right] \right]_{\pi/2}^{\pi} \right\}$$

$$= \frac{2k}{\pi} \left\{ \left[-\frac{\pi}{2} \frac{\cos \frac{n\pi}{2}}{n} + \frac{1}{n^2} \sin \frac{n\pi}{2} \right] - \left[0 - 0 - \frac{\pi}{2} \frac{\cos \frac{n\pi}{2}}{n} - \frac{\sin \frac{n\pi}{2}}{n^2} \right] \right\}$$

$$= \frac{2k}{\pi} \cdot \frac{2}{n^2} \sin \frac{n\pi}{2}$$

$$= \frac{4k}{\pi n^2} \sin \frac{n\pi}{2}$$

$\therefore$ Fourier sine series is

$$f(x) = \sum_{n=1}^{\infty} \frac{4k}{\pi n^2} \sin \frac{n\pi}{2} \sin nx$$

$$= \frac{4k}{\pi} \left[\frac{\sin x}{1^2} - \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} - \dots \right]$$

20 a) Find the Fourier cosine series of $f(x) = x^2$ in $(0, \pi)$. Hence show that $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$

Fourier cosine series of $f(x)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{l} \int_0^l f(x) dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x^2 dx$$

$$= \frac{1}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi} = \frac{1}{\pi} \times \frac{\pi^3}{3} = \frac{\pi^2}{3}$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx) dx$$

$$= \frac{2}{\pi} \left[x^2 \times \frac{\sin(n\pi)}{n} - 2x \times \frac{-\cos(n\pi)}{n^2} + 2 \times \frac{-1}{n^3} \times \sin(n\pi) \right]$$

$$= \frac{2}{\pi} \left[\frac{2\pi}{n^2} \cos n\pi \right] = \frac{4}{n^2} (-1)^n$$

$$\therefore f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos nx$$

$$f(x) = \frac{\pi^2}{3} + \frac{4}{1^2} \times -1 \cdot \cos x + \frac{4}{2^2} \cos 2x + \frac{4}{3^2} \times -1 \times \cos 3x + \dots$$

$$x^2 = \frac{\pi^2}{3} + 4 \left[-\cos x + \frac{1}{2^2} \cos 2x - \frac{1}{3^2} \cos 3x + \dots \right]$$

When $x = \pi$,

$$\pi^2 = \frac{\pi^2}{3} + 4 \left[-\cos \pi + \frac{1}{2^2} \cos 2\pi - \frac{1}{3^2} \cos 3\pi + \dots \right]$$

$$\pi^2 = \frac{\pi^2}{3} + 4 \left[1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right]$$

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{1}{4} \left[\pi^2 - \frac{\pi^2}{3} \right]$$

$$= \frac{1}{4} \left[\frac{2\pi^2}{3} \right]$$

$$= \frac{\pi^2}{6}$$

$$\therefore 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$$

20 b) Find the fourier series for the function

$$f(x) = \begin{cases} x, & 0 < x < 1 \\ 1-x, & 1 < x < 2 \end{cases}$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{2l} \int_{\alpha}^{\alpha+2l} f(x) dx$$

$$= \frac{1}{2} \left[\int_0^1 x dx + \int_1^2 (1-x) dx \right]$$

$$= \frac{1}{2} \left[\left[\frac{x^2}{2} \right]_0^1 + \left[x - \frac{x^2}{2} \right]_1^2 \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} + 2 - \frac{4}{2} - \left(1 - \frac{1}{2}\right) \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} - 1 + \frac{1}{2} \right]$$

$$= 0$$

$$a_n = \frac{1}{l} \int_{\alpha}^{\alpha+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \int_0^1 x \cos(n\pi x) dx + \int_1^2 (1-x) \cos(n\pi x) dx$$

$$= x \times \frac{\sin(n\pi x)}{n\pi} - 1 \times \frac{1}{n^2 \pi^2} x^{-\cos(n\pi x)} \Big|_0^1 + (1-x) \cdot \frac{\sin(n\pi x)}{n\pi} - 1 \times \frac{-1}{n^2 \pi^2} \cos(n\pi x) \Big|_1^2$$

$$= \frac{1}{n\pi} \times \sin n\pi + \frac{1}{n^2 \pi^2} \cos n\pi - \left(\frac{1}{n^2 \pi^2}\right) + \frac{-1}{n^2 \pi^2} \times 1 - \left[\left(\frac{-1}{n^2 \pi^2} \times (-1)^n\right)\right]$$

$$= \frac{1}{n^2 \pi^2} 2(-1)^n - \frac{1}{n^2 \pi^2} - \frac{1}{n^2 \pi^2} + \frac{1}{n^2 \pi^2} (-1)^n$$

$$= 2 \times \frac{1}{n^2 \pi^2} (-1)^m - 2 \times \frac{1}{n^2 \pi^2}$$

$$= \frac{2}{n^2 \pi^2} [(-1)^m - 1]$$

$$b_m = \frac{1}{l} \int_a^{a+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \int_0^1 x \sin(n\pi x) dx + \int_1^2 (1-x) \sin(n\pi x) dx$$

$$= x \times \frac{-\cos(n\pi x)}{n\pi} - 1 \times \frac{-1}{n^2 \pi^2} \sin(n\pi x) \Bigg|_0^1$$

$$+ (1-x) \cdot \frac{-\cos(n\pi x)}{n\pi} - -1 \times \frac{-1}{n^2 \pi^2} \sin(n\pi x) \Bigg|_1^2$$

$$= \frac{-1}{n\pi} (-1)^m + \frac{1}{n\pi} \times 1$$

$$= \frac{1}{n\pi} [1 - (-1)^m]$$

$$f(x) = \frac{2}{n^2 \pi^2} [(-1)^m - 1] \cos(n\pi x) + \frac{1}{n\pi} [1 - (-1)^m] \sin(n\pi x)$$